

M³: A Sub-Millisecond Scheduler for Multi-Cell MIMO Networks under C-RAN Architecture

Yongce Chen[†] Y. Thomas Hou[†] Wenjing Lou[†] Jeffrey H. Reed[†] Sastry Kompella[‡]

[†]Virginia Tech, Blacksburg, VA, USA

[‡]U.S. Naval Research Laboratory, Washington, DC, USA

Abstract—Cloud Radio Access Network (C-RAN) is a novel centralized architecture for cellular networks. C-RAN can significantly improve spectrum efficiency by performing cooperative signal processing for multiple cells at a centralized baseband unit (BBU) pool. However, a new resource scheduler is needed before we can take advantage of C-RAN's multi-cell processing capability. Under C-RAN architecture, the scheduler must jointly determine RB allocation, MCS assignment, and beamforming matrices for all users under all covering cells. In addition, it is necessary to obtain a scheduling solution within each TTI (at most 1 ms) to be useful for the frame structure defined by 5G NR. In this paper, we present M³—a sub-ms scheduler for multi-cell MIMO networks under C-RAN architecture. M³ addresses the stringent timing requirement through a novel multi-pipeline design that exploits parallelism. Under this design, one pipeline performs a sequence of operations for cell-edge users to explore joint transmission, and in parallel, the other pipeline is for cell-center users to explore MU-MIMO transmission. Experimental results show that M³ is capable of offering a scheduling solution within 1 ms for 7 remote radio heads (RRHs), 100 users, 100 RBs, and 2×12 MIMO. Meanwhile, M³ provides ~40% throughput gain on average by employing joint transmission.

I. INTRODUCTION

To increase spectrum efficiency and reduce the operation cost for the next-generation cellular systems, the so-called “C-RAN architecture” has been explored [1–5]. As shown in Fig. 1, C-RAN is based on a centralized architecture—a baseband unit (BBU) pool located at a centralized site serving several remote radio heads (RRHs). The BBU pool is responsible for the data processing at upper PHY layer (i.e., baseband signal processing), MAC layer and network layer for all RRHs under its coverage. Each RRH is equipped with multiple antennas—they are responsible for lower PHY layer signal processing, i.e., performing the radio frequency functions and emitting the signals.¹ The connection between the BBU pool and the remote RRHs is through high-capacity, low-latency optical fronthauls. Per 5G specifications, the maximum allowed end-to-end one-way latency of functional split between upper PHY and lower PHY is 250 μ s [7], which is sufficiently small compared with the channel coherence time for most communication scenarios. Therefore, the real-time scheduling can be performed at the center BBU pool.

A goal of C-RAN is to support *joint transmission*—a coordinated beamforming scheme that can significantly improve

¹The point of separation between BBU pool and RRHs in functionalities may vary and depends on different options in 5G [6].

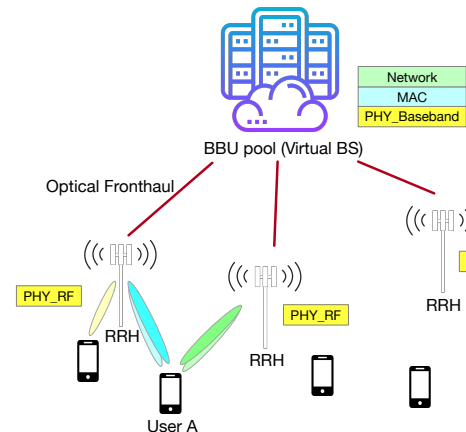


Fig. 1: Under C-RAN architecture, a centralized BBU pool is scheduling resources for users covered by a set of RRHs. A user can receive its data from one or multiple RRHs at the same time.

spectrum efficiency [1–3, 7, 8]. By “joint transmission”, we mean a user can receive data from multiple RRHs simultaneously (see user A in Fig. 1). As future networks become smaller and denser, there is an opportunity for joint transmission. Likewise, effective management of inter-cell interference becomes even more important. Thanks to centralized architecture in C-RAN, the virtual BS (i.e., BBU pool) can ease the sharing of signaling, traffic data and channel state information that are needed for joint transmission from different cells.

However, significant challenges remain in the design of a C-RAN scheduler for joint transmission. Considering a downlink scheduling problem in C-RAN, we face the following critical challenges. First, the centralized scheduler must allocate a number of resource blocks (RBs) and decide the beamforming matrices at each RRH. Under joint transmission, the solutions must be done jointly across different cells (covered by different RRHs) at the virtual BS in each TTI. Second, the scheduler must assign MCS and number of data streams for each user. Under 5G NR [9], a user’s receiver must have the same MCS level and number of data streams across all RBs that are allocated to her. Third, to achieve high throughput, MU-MIMO transmission should be used under C-RAN. Thus, an RB may be allocated to multiple users.

In addition to the above challenges on the scheduler side,

we also have a stringent timing requirement—the scheduler must find its scheduling solution within one TTI. Under 5G numerology 0, one TTI is 1 ms. Then the C-RAN scheduling solution must be found within 1 ms to be useful. To support ultra-low latency applications, an even shorter TTI may be needed (e.g., 500 μ s under numerology 1).

To date, there has been a number of studies on scheduling or beamforming problems under C-RAN (see, e.g., [10–17]). However, none of these studies considered the real-time requirement. For example, the authors in [10–14] designed coordinated beamforming schemes for C-RAN, but these designs are based on iterative optimization, which requires an excessive amount of computation time. Further, none of these works jointly optimizes RB allocation, MCS assignment and beamforming matrices for a multi-cell system. For the studies in [10–17], although coordinated beamforming is considered, either RB allocation or MCS assignment is missing.

In this paper, we present the design and implementation of M^3 —the first sub-Millisecond scheduler for Multi-cell MIMO networks under 5G C-RAN architecture. M^3 is capable of finding a solution to RB allocation, MCS selection, data stream assignment, as well as the precoding matrices, in real-time for each TTI (at most 1 ms). We tackle the crucial timing problem through a novel parallel design on a commercial off-the-shelf (COTS) GPU platform. Our main contributions can be summarized as follows:

- M^3 is the first C-RAN scheduler that can meet the 1 ms real-time requirement. The success of M^3 is built upon a judicious parallel design and validated on a COTS GPU. This design is developed in accordance with the time-frequency resource structure defined by 5G NR, and it is applicable to centralized multi-cell systems.
- M^3 exploits independency and parallelism through a multi-pipeline design. Specifically, we first divide all users into two groups: non-edge users and cell-edge users by leveraging their channel properties. Then M^3 performs two independent parallel pipelines, with one pipeline focusing on a sequence of operations for cell-edge users and the other pipeline for non-edge users. After both pipelines complete their operations in parallel, we determine the final solutions for all users.
- M^3 achieves large-scale parallelism in addition to the multi-pipeline structure. Throughout our design, the exploration of parallel computing is carried out by leveraging GPU's capability. For instance, within each pipeline, most operations are designed to be independent among RRHs and RBs. By taking advantage of massive parallel computation all the way, our design can reduce the computation time dramatically.
- M^3 is implemented on a COTS GPU platform and extensive experiments are conducted to verify its performance. Our experimental results show that M^3 is able to offer the scheduling solution within 1 ms for a C-RAN system with up to 7 RRHs, 100 users, 100 RBs, and 2×12 MIMO. In the meantime, M^3 achieves $\sim 40\%$ throughput gain under joint transmission.

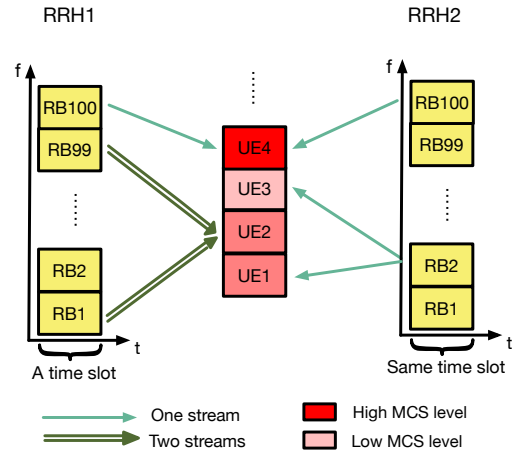


Fig. 2: Within each time slot, the virtual BS jointly determines RB allocation, number of data streams, and MCS assignment for all users under all RRHs.

II. SYSTEM MODEL

We consider a downlink (DL) scheduling problem under C-RAN architecture. As shown in Fig. 1, a centralized BBU pool is connected to a set $\mathcal{L}$ of RRHs (i.e., $|\mathcal{L}|$ cells). This BBU pool serves a set $\mathcal{K}$ of users. Each RRH is equipped with N_T antennas while each user is equipped with N_R antennas and $N_T > N_R$. Fig. 2 illustrates our scheduling problem, which we elaborate mathematically in the rest of this section.

User Association and RB Allocation Consider a frequency reuse system, where a wide frequency band is reused at every RRH. Per 3GPP, the frequency band is divided into a set $\mathcal{B}$ of RBs. In each TTI, $\mathcal{B}$ is available at each RRH for DL transmission. Under C-RAN architecture, a user can receive its signals from one or multiple RRHs on an RB. Under joint transmission [3, 8], when user k is receiving its signals on RB b from multiple RRHs, the user data s_k^b is identical from these RRHs. Denote $x_{l,k}^b(t) \in \{0, 1\}$ as a binary variable indicating whether or not RRH l is transmitting data to user k on RB b in TTI t , i.e.,

$$x_{l,k}^b(t) = \begin{cases} 1, & \text{if RRH } l \text{ is transmitting data} \\ & \text{to user } k \text{ on RB } b \text{ in TTI } t, \\ 0, & \text{otherwise.} \end{cases}$$

Denote $\mathcal{L}_k^b(t)$ as the set of RRHs that are transmitting data to user k on RB b in TTI t , i.e., $\mathcal{L}_k^b(t) = \{l \in \mathcal{L} | x_{l,k}^b(t) = 1\}$. Denote $\hat{x}_k^b(t) \in \{0, 1\}$ as a binary variable indicating whether or not user k is receiving data from at least one RRH on RB b in TTI t , i.e.,

$$\hat{x}_k^b(t) = \begin{cases} 1, & \text{if } |\mathcal{L}_k^b(t)| > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Also, note that under MU-MIMO, an RRH can transmit to multiple users on the same RB. As the maximum number of users scheduled on an RB cannot exceed the number of

antennas at the RRH, we have the following MU-MIMO constraint for $x_{l,k}^b(t)$:

$$\sum_{k \in \mathcal{K}} x_{l,k}^b(t) \leq N_T. \quad (b \in \mathcal{B}, l \in \mathcal{L}) \quad (2)$$

Number of Data Streams A user may have multiple data streams on each RB that is allocated to her. But when a user receives its data streams on multiple RBs, then the number of data streams must be identical across all these RBs [18].

Denote $y_k(t)$ as the number of data streams for user k in TTI t (which is the same across all allocated RBs). As $y_k(t)$ cannot be greater than the number of receive antennas, i.e.,

$$y_k(t) \leq N_R. \quad (k \in \mathcal{K}) \quad (3)$$

Also, at each RRH, the total number of data streams on each RB for MU-MIMO transmission cannot exceed the number of its antennas. We have

$$\sum_{k \in \mathcal{K}} x_{l,k}^b(t) y_k(t) \leq N_T. \quad (b \in \mathcal{B}, l \in \mathcal{L}) \quad (4)$$

Achieved SINR at Users Each RRH applies precoders to support joint transmission and/or MU-MIMO transmission. Let $\mathbf{F}_{l,k}^b(t)$ be an $N_T \times x_{l,k}^b(t) y_k(t)$ precoding matrix for user k used by RRH l on RB b . Under the power constraint at an RRH, we have $\sum_{k \in \mathcal{K}} \|\mathbf{F}_{l,k}^b(t)\|_F^2 \leq P_T$ for all $l \in \mathcal{L}$, where P_T is the total power (per RB) at the RRH and $\|\cdot\|_F$ denotes the Frobenius norm. Then the received signal of user k on RB b is given by

$$\mathbf{c}_k^b = \sum_{l \in \mathcal{L}_k^b} \mathbf{H}_{l,k}^b \mathbf{F}_{l,k}^b \mathbf{s}_k^b + \sum_{l \in \mathcal{L} \setminus \mathcal{L}_k^b} \mathbf{H}_{l,k}^b \mathbf{F}_{l,i}^b \mathbf{s}_i^b + \mathbf{n}_k^b,$$

where $\mathbf{H}_{l,k}^b \in \mathbb{C}^{N_R \times N_T}$ is the channel matrix from RRH l to user k on RB b , $\mathbf{n}_k^b$ is the $N_R \times 1$ vector of i.i.d $\mathcal{CN}(0, n_0^2)$ additive complex Gaussian noise, $\mathbf{s}_k^b$ is the signal vector. We omit the time-dependent notation (t) for matrices for brevity.

Then each user applies an $N_R \times y_k(t)$ combiner $\mathbf{W}_k^b$ for the received signals. After this combiner, we have the following signal for user k on RB b :

$$\begin{aligned} \tilde{\mathbf{c}}_k^b &= \mathbf{W}_k^{b\dagger} \mathbf{c}_k^b \\ &= \underbrace{\sum_{l \in \mathcal{L}_k^b} \mathbf{W}_k^{b\dagger} \mathbf{H}_{l,k}^b \mathbf{F}_{l,k}^b \mathbf{s}_k^b}_{\text{desired signal}} + \underbrace{\sum_{l \in \mathcal{L} \setminus \mathcal{L}_k^b} \sum_{i \in \mathcal{K}} \mathbf{W}_k^{b\dagger} \mathbf{H}_{l,k}^b \mathbf{F}_{l,i}^b \mathbf{s}_i^b}_{\text{interference}} + \mathbf{W}_k^{b\dagger} \mathbf{n}_k^b, \end{aligned}$$

where $(\cdot)^\dagger$ denotes the conjugate transpose of a matrix. Different beamforming schemes can be applied based on $\mathbf{H}_{l,k}^b$. In this paper, we apply MMSE beamforming scheme at RRHs and users with equal power allocation.

For each $k \in \mathcal{K}$, the signal-to-interference-plus-noise ratio (SINR) of the f -th stream on RB b is then given by

$$\text{SINR}_k^{b,f} = \frac{E_k^{b,f}}{\mathbf{Q}_k^{b,f} + n_0^2 \mathbf{W}_k^{b,f\dagger} \mathbf{W}_k^{b,f}}, \quad (5)$$

where

$$\begin{aligned} E_k^{b,f} &= \left| \sum_{l \in \mathcal{L}_k^b} \mathbf{W}_k^{b,f\dagger} \mathbf{H}_{l,k}^b \mathbf{F}_{l,k}^{b,f} \right|^2 \\ \mathbf{Q}_k^b &= \sum_{i \in \mathcal{K}} \sum_{f'=1}^{y_i(t)} \left| \sum_{l \in \mathcal{L}} \mathbf{W}_k^{b,f\dagger} \mathbf{H}_{l,k}^b \mathbf{F}_{l,i}^{b,f'} \right|^2 - E_k^{b,f}, \end{aligned}$$

and $(\cdot)_k^{b,f}$ is the f -th column of $(\cdot)_k^b$.

MCS Assignment In each TTI t , a set $\mathcal{M}$ of MCSs is available for data transmission for each user $k \in \mathcal{K}$. However, if a user is scheduled to receive data streams on multiple RBs, 3GPP requires that the user employs the same MCS across all data streams on all scheduled RBs [9]. Denote $z_k^m(t) \in \{0, 1\}$ as a binary variable indicating whether or not MCS $m \in \mathcal{M}$ is used by the virtual BS for user $k \in \mathcal{K}$ in TTI t , i.e.,

$$z_k^m(t) = \begin{cases} 1, & \text{if MCS } m \text{ is used for user } k \text{ in TTI } t, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

To guarantee only one MCS is used across all scheduled RBs for user k , we have

$$\sum_{m \in \mathcal{M}} z_k^m = 1. \quad (k \in \mathcal{K}) \quad (7)$$

Instantaneous Data Rate The instantaneous achievable data rate depends on the SINR of each stream and the selected MCS level. Specifically, with a higher MCS level m , the user data is encoded with a higher data rate r^m . However, to successfully decode the user data at a higher MCS level m , a higher level of SINR is required, or the data cannot be successfully decoded (i.e., a data rate of zero). Denote θ^m as the SINR threshold for successfully decoding the data with MCS m , and $r_k^{b,f,m}(t)$ as the instantaneous achievable data rate of user k 's f -th data stream on RB b with MCS m in TTI t . Then we have

$$\begin{aligned} r_k^{b,f,m}(t) &= \begin{cases} r^m, & \text{if } \text{SINR}_k^{b,f} \geq \theta^m, \\ 0, & \text{otherwise.} \end{cases} \\ (f &= 1, \dots, y_k(t), k \in \mathcal{K}, b \in \mathcal{B}, m \in \mathcal{M}) \end{aligned} \quad (8)$$

where $\text{SINR}_k^{b,f}$ is defined in Eq. (5).

The aggregate achievable data rate of user k in TTI t can be given by

$$R_k(t) = \sum_{b \in \mathcal{B}} \hat{x}_k^b(t) \sum_{f=1}^{y_k(t)} \sum_{m \in \mathcal{M}} z_k^m(t) r_k^{b,f,m}(t), \quad (9)$$

where $\hat{x}_k^b(t)$ is defined in Eq. (1), and we define $\sum_{f=1}^{y_k(t)} (\cdot) = 0$ if $y_k(t) = 0$.

Proportional Fair Metric A common scheduling objective is to maximize the PF metric. Specifically, a PF-oriented scheduler aims at maximizing the utility function $\sum_{k \in \mathcal{K}} \log \bar{R}_k$, where $\bar{R}_k$ is the long-term average data rate of user k . For a time-slotted system, a widely used approach is to maximize the sum of normalized data rates in each time slot [17, 19, 20],

$$\sum_{k \in \mathcal{K}} \frac{R_k(t)}{\bar{R}_k(t-1)}, \quad (10)$$

where the long-term average data rates $\tilde{R}_k(t-1)$ are updated using an exponentially weighted filter:

$$\tilde{R}_k(t-1) = \frac{T_c - 1}{T_c} \tilde{R}_k(t-2) + \frac{1}{T_c} R_k(t-1).$$

Problem Statement Our objectives are 1) to determine the users' RRH association and allocate RBs ($x_{l,k}^b(t)$'s), assign the number of data streams ($y_k(t)$'s) and MCSs ($z_k^m(t)$'s), as well as compute precoding matrices ($\mathbf{F}_{l,k}^b(t)$'s) for all users, such that the PF metric (10) is maximized; and 2) to ensure the scheduling solution can be found within each TTI (i.e., at most 1 ms) to meet 5G's timing requirement. This C-RAN scheduling problem can be written as follows.

OPT

$$\begin{aligned} \max \quad & \sum_{b \in \mathcal{B}} \sum_{k \in \mathcal{K}} \sum_{m \in \mathcal{M}} \sum_{f=1}^{y_k(t)} \frac{r_k^{b,f,m}(t)}{\tilde{R}_k(t-1)} \hat{x}_k^b(t) z_k^m(t), \\ \text{s.t.} \quad & \text{User association and RB allocation constraint: (2);} \\ & \text{Data stream allocation constraints: (4);} \\ & \text{MCS assignment constraint: (7);} \\ & \text{SINR and instantaneous data rate: (5), (8);} \\ & x_{l,k}^b(t) \in \{0, 1\}, y_k(t) \in \{0, 1, \dots, N_R\}, z_k^m(t) \in \{0, 1\}. \end{aligned}$$

In problem OPT, $x_{l,k}^b(t)$, $y_k(t)$ and $z_k^m(t)$ are decision variables, $\mathbf{F}_{l,k}^b(t)$, $\mathbf{W}_k^b(t)$, $\hat{x}_k^b(t)$, $r_k^{b,f,m}(t)$ and $\tilde{R}_k(t-1)$ are intermediate variables which can be determined with given $x_{l,k}^b(t)$, $y_k(t)$ and $z_k^m(t)$. The others are given constants.

Problem OPT is a nonlinear integer problem. The decision variables are tightly coupled together with an extremely large search space. Further, the stringent timing requirement adds another level of challenge to design a solution for OPT.

III. $\mathbf{M}^3$: KEY IDEAS AND ROADMAP

Before we present the design blueprint of $\mathbf{M}^3$, let's first offer some insights on the C-RAN scheduling problem. Different from traditional single-cell resource scheduling, C-RAN architecture has the potential to further improve PF objective through cooperative scheduling among neighboring cells. For cooperative scheduling, we first need to address the critical question of how to allocate RBs at each RRH for potential joint transmission. We have the following two options:

- (i) Multiple RRHs use this RB to form joint transmission to a user;
- (ii) Each RRH uses this RB separately for different users.

There is a trade-off between the above two options. On the one hand, if an RB is used for joint transmission at multiple RRHs, potential interfering signals are effectively transformed into desired signals, which can significantly improve a receiver's data rate. On the other hand, joint transmission consumes more RB resources—a user that is supported by joint transmission requires all cooperative RRHs to use this RB for her. Without joint transmission, these RRHs may use this RB separately (and independently) to support more users.

$\mathbf{M}^3$ tackles the RB allocation problem with the following ideas. First, it is clear that not every user can benefit from joint

transmission. Under a virtual BS, we have multiple adjacent RRHs, forming multiple adjacent cells in the system. Then only cell-edge users may benefit from receiving signals from multiple RRHs, while other (non-edge) users are better served solely by their own RRHs. Therefore, $\mathbf{M}^3$ reduces the search space by dividing all users into two groups: cell-edge users $\mathcal{K}^E$ and non-edge users $\mathcal{K}^N$. Only users in $\mathcal{K}^E$ have the opportunity for joint transmission. This division should be based on large-scale path loss and can be determined based on long-term measurements.

Second, based on above user division, $\mathbf{M}^3$ exploits parallelism to determine RB allocation, along with MCS and number of data streams, to problem OPT. Specifically, $\mathbf{M}^3$ employs two independent *parallel pipelines*, where one pipeline is a sequence of operations focusing on cell-edge users (to explore joint transmission) and the other pipeline is for non-edge users (to explore MU-MIMO transmission). To determine whether an RB b is better used for $\mathcal{K}^E$ or $\mathcal{K}^N$ in terms of maximizing the PF metric, $\mathbf{M}^3$ will check both cases in parallel by using those two pipelines. That is, $\mathbf{M}^3$ allows both pipelines to use each RB, determines the scheduling solutions by each pipeline (for $\mathcal{K}^E$ or $\mathcal{K}^N$), and then obtains the PF values achieved on each RB in both cases by these pipelines. Then, by comparing the PF values obtained by each pipeline, $\mathbf{M}^3$ makes the final decision on how each RB is allocated, as well as corresponding assignment of MCS and number of data streams. Note that in addition to the multi-pipeline structure, parallelism is carried out throughout the design within each pipeline. For example, many operations are purposely designed to be independent among RRHs and RBs, and therefore they can be implemented in parallel.

Fig. 3 shows a flow chart of $\mathbf{M}^3$, which consists of the following key steps.

- Stage I: User classification. $\mathbf{M}^3$ divides all users into two groups: cell-edge users $\mathcal{K}^E$ and non-edge users $\mathcal{K}^N$, based on large-scale path loss. For each user k , $\mathbf{M}^3$ identifies its potential serving set $\mathcal{L}_k$.
- Stage II: Find promising solutions of RB allocation, as well as MCS assignment and number of data streams, for both cell-edge and non-edge users, by performing two independent parallel pipelines:
 - P1: Cell-Edge Pipeline. P1 finds promising solutions for $k \in \mathcal{K}^E$, assuming each RB is used for cell-edge users.
 - P2: Non-Edge Pipeline. P2 finds promising solutions for $k \in \mathcal{K}^N$, assuming each RB is used for non-edge users.
- Stage III: Compare the PF metrics provided by each pipeline and determine the final solution.

IV. $\mathbf{M}^3$: DESIGN DETAILS

In this section, we described the operations of $\mathbf{M}^3$ in detail at each stage.

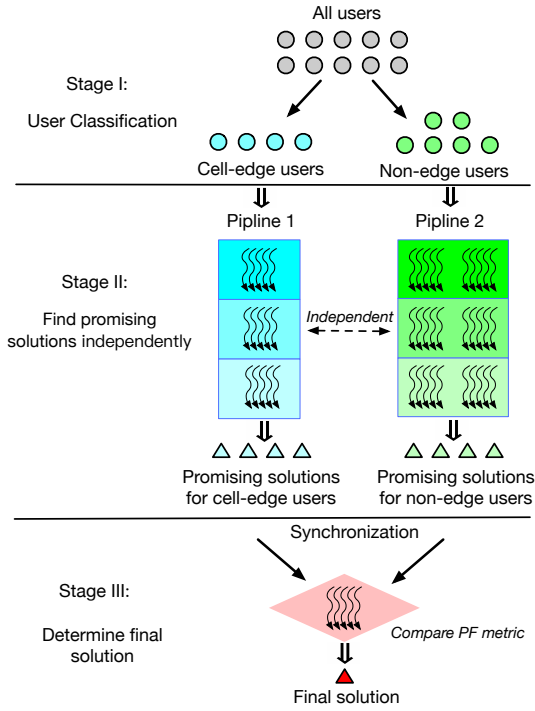


Fig. 3: A flow chart for M^3 .

A. Stage I: User classification

In this stage, we divide all users into two groups: cell-edge users $\mathcal{K}^E$ and non-edge users $\mathcal{K}^N$. For non-edge users, they are likely to experience much better channel qualities from their closest RRH than other RRHs. Therefore, users identified in $\mathcal{K}^N$ will only be served by their closest RRH. Also, an RRH may perform MU-MIMO transmission to multiple non-edge users under this RRH to achieve higher throughput. On the other hand, for cell-edge users, they are likely to experience similar signal strength from at least two RRHs. To reinforce the desired signal strength, users identified in $\mathcal{K}^E$ will be served by multiple RRHs simultaneously. M^3 only applies SU-MIMO transmission for cell-edge users, as MU-MIMO transmission is not beneficial for improving each individual user's SINR, which is what a cell-edge user needs.

M^3 determines whether a user is a non-edge user or a cell-edge user, as well as its RRH(s), based on the relative large-scale path loss from different RRHs to this user. Fast fading and beamforming gain are not considered in this stage. Let $g_{l,k}$ be the path loss from the RRH l to user k . Then the set of user k 's RRH(s) is given by

$$\mathcal{L}_k = \left\{ l \in \mathcal{L} \left| \frac{g_{l,k}}{\min_{n \in \mathcal{L}} g_{n,k}} \leq \delta \right. \right\}, \quad (k \in \mathcal{K})$$

where δ ($\delta \geq 1$) is a pre-defined threshold to determine the subset of RRHs.

Next, if $\mathcal{L}_k$ for user k has more than one RRH, then user k is classified as a cell-edge user. Denote $\mathcal{K}^E$ as the set of all cell-edge users. Then $\mathcal{K}^E = \{k | |\mathcal{L}_k| > 1, k \in \mathcal{K}\}$. Otherwise, if $\mathcal{L}_k$ for user k has only one RRH, then user k is classified as

a non-edge user. Denote $\mathcal{K}^N$ as the set of all non-edge users. Then $\mathcal{K}^N = \{k | |\mathcal{L}_k| = 1, k \in \mathcal{K}\}$. Further, denote $\mathcal{K}_l^N$ as the subset of non-edge users in $\mathcal{K}^N$ that are receiving service from RRH l . Then $\mathcal{K}_l^N = \{k | \mathcal{L}_k = \{l\}, k \in \mathcal{K}^N\}$.

Since the decision of user-RRH association largely depends on the user's location, $\mathcal{L}_k$ can be updated based on long-term measurement. Therefore, Stage I will not be counted toward M^3 's total execution time.

Upon the completion of Stage I, the search space for $x_{l,k}^b(t)$ variables will be narrowed down. That is, we have $x_{l,k}^b(t) = 0$ if $l \notin \mathcal{L}_k$. Stage I lays the foundation for employing independent parallel pipelines, which we will describe in Stage II.

B. Stage II: Find promising solutions for cell-edge and non-edge users

Stage II consists of two independent pipelines that can be implemented in parallel. Each pipeline is a sequence of operations to find promising solutions for users in $\mathcal{K}^E$ and $\mathcal{K}^N$, respectively. Within each pipeline, we have three key steps. Each step focuses on one type of variable (i.e., $x_{l,k}^b(t)$'s, $y_k(t)$'s or $z_k^m(t)$'s) for $\mathcal{K}^E$ or $\mathcal{K}^N$. That is, in each step, M^3 will restrict the search space for that type of variable into a small but promising subspace. As RB allocation is the key problem for joint transmission under C-RAN, we start with $x_{l,k}^b(t)$ variables first under both pipelines. Then it is followed by steps for $y_k(t)$ and $z_k^m(t)$ variables.

Pipeline 1: Edge-Users It consists of the following two steps.

Step 1-A: Determine RB Allocation for Cell-Edge Users. In this step, we focus on the $x_{l,k}^b(t)$ variable for cell-edge users. That is, M^3 identifies a subset $\mathcal{B}_k^E$ of promising RBs from $\mathcal{B}$ for each user $k \in \mathcal{K}^E$. Note that all RRHs for user k (i.e., all $l \in \mathcal{L}_k$) must use the same RBs to perform joint transmission to user k .

To maximize the PF objective function (10), it is equivalent to maximize each user's instantaneous data rate normalized by its long-term data rate (i.e., $\frac{R_k(t)}{\tilde{R}_k(t-1)}$). For a cell-edge user under joint transmission, the instantaneous data rate is tightly related to the aggregated channel quality $(\sum_{l \in \mathcal{L}_k} \sqrt{P_T} \|\mathbf{H}_{l,k}^b\|_F)^2$. Intuitively, when user k is experiencing high channel qualities from all its RRHs on an RB b , scheduling this RB jointly by $\mathcal{L}_k$ is likely to achieve a high data rate for this user. The achievable data rate can be approximated based on the channel capacity formula, i.e., $\log_2 \left(1 + \left(\sum_{l \in \mathcal{L}_k} \sqrt{P_T} \|\mathbf{H}_{l,k}^b\|_F \right)^2 \frac{1}{n_0^2} \right)$. Further, the data rate should be normalized by the user's long-term average data rate $\tilde{R}_k(t-1)$ in order to maximize the PF objective. Therefore, we consider the following metric to determine $\mathcal{B}_k^E$:

$$q_k^{b,E} = \frac{\log_2 \left(1 + \left(\sum_{l \in \mathcal{L}_k} \sqrt{P_T} \|\mathbf{H}_{l,k}^b\|_F \right)^2 \frac{1}{n_0^2} \right)}{\tilde{R}_k(t-1)},$$

where $\|\mathbf{H}_{l,k}^b\|_F$ is the Frobenius norm of $\mathbf{H}_{l,k}^{b,E}$.

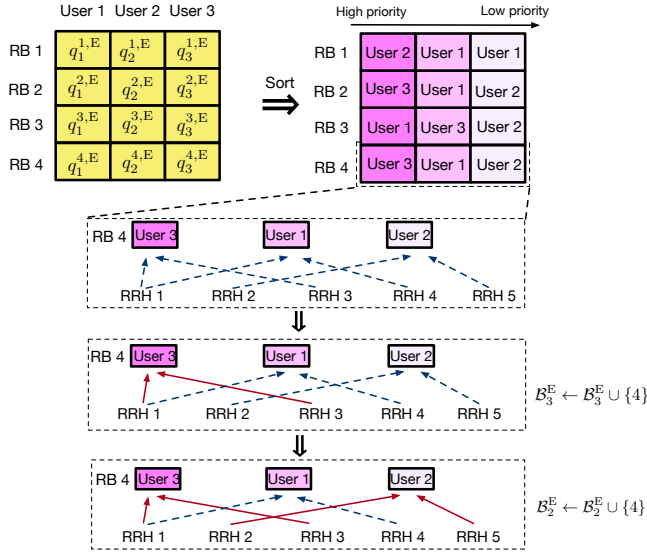


Fig. 4: An illustration of Step 1-A in Pipeline 1.

Fig. 4 illustrates how $\mathbf{M}^3$ finds $\mathcal{B}_k^E$. First, $\mathbf{M}^3$ creates $|\mathcal{B}| |\mathcal{K}^E|$ independent parallel flows, each of which calculates one $q_k^{b,E}$ for a user k and RB b . Second, $\mathbf{M}^3$ generates $|\mathcal{B}|$ independent flows, where each flow sorts $\{q_1^{b,E}, q_2^{b,E}, \dots, q_{|\mathcal{K}^E|}^{b,E}\}$ in descending order. The sorting result indicates each user's priority to use an RB b . Third, with $|\mathcal{B}|$ independent flows, $\mathbf{M}^3$ allocates each RB based on the priority list and resolves the potential conflicts among RRHs. We describe the conflict resolution process through a simple example (see the bottom half in Fig. 4). For RB 4 in Fig. 4, RRHs 1 and 3 allocate RB 4 to user 3 because user 3 has the highest priority on this RB. Subsequently, user 1 cannot have RB 4, as RRH 1—one of its RRHs, has reserved this RB for user 3. But user 2 can have RB 4 because none of its RRHs have allocated RB 4 to a user with a priority higher than user 2.

Now we have the RB allocation results $\mathcal{B}_k^E$ for each cell-edge user, which restricts the decision of $x_{l,k}^{b,E}(t)$ variables. That is, if $k \in \mathcal{K}^E$ and $b \notin \mathcal{B}_k^E$, then we have $x_{l,k}^{b,E}(t) = 0$ for all $l \in \mathcal{L}_k$. For $k \in \mathcal{K}^E$ and $b \in \mathcal{B}_k^E$, a potential (promising) solution is to let $x_{l,k}^{b,E}(t) = 1$. But the final decision will be made later (after we compare the solution from the non-edge pipeline).

Step 1-B: Determine Number of Data Streams for Cell-Edge Users. In this step, we focus on the $y_k(t)$ variables for cell-edge users. There exists a trade-off between the achievable data rate and number of data streams for a user. On one hand, more data streams have the potential to increase a user's total data rate, leveraging the spatial multiplexing for MIMO channels. On the other hand, transmitting too many data streams may cause performance loss, due to power splitting and interference among the streams.

To gain some insights on how we should decide the value of $y_k(t)$, let's first consider a single-link MIMO channel $\mathbf{H}$. Consider the eigenmode beamforming and equal power allocation. The achievable data rate can be given as a function

of the number of data streams [22], i.e.,

$$r(y) = \sum_{i=1}^y \log \left(1 + \frac{P_T [\sigma(i)]^2}{n_0^2} \right), \quad (11)$$

where y is the number of data streams on channel $\mathbf{H}$ ($y \leq \text{rank}(\mathbf{H})$) and $\sigma(i)$ is the i -th largest eigenvalue of $\mathbf{H}$. Eq. (11) shows the fundamental relationship between achievable data rate and the number of streams on a single-link MIMO channel, which is tightly related to per-stream SNR ($\frac{P_T}{y n_0^2}$) and singular values $\sigma(i)$.

Inspired by Eq. (11), we propose the following approach to determine the value of $y_k(t)$ in the context of joint transmission. First, recall that a user must use the same MCS across all streams. This suggests that we should consider the lowest SINR among user k 's all streams. If user k is receiving f streams, then we evaluate an estimated SINR of the stream with the f -th largest eigenvalue, which is defined as

$$\gamma_k^E(f) = \frac{\left(\sum_{l \in \mathcal{L}_k} \sqrt{\frac{P_T}{f}} E_b[\sigma_{l,k}^{b,E}(f)] \right)^2}{n_0^2}.$$

In the above expression, $\sigma_{l,k}^{b,E}(f)$ is the f -th largest eigenvalue of $\mathbf{H}_{l,k}^b$ and $E_b[\sigma_{l,k}^{b,E}(f)]$ is the average eigenvalue over all $b \in \mathcal{B}_k^E$. Obviously, we have $\gamma_k^E(1) > \gamma_k^E(2) > \dots \geq \gamma_k^E(N_R)$ for any user k .

Next, let $\bar{m}_k^E(f)$ be the largest MCS level in $\mathcal{M}$ that can be used to satisfy user k 's estimated SINR $\gamma_k^E(f)$, i.e.,

$$\bar{m}_k^E(f) = \max_{m \in \mathcal{M}} m \quad \text{s.t. } \gamma_k^E(f) \geq \theta^m.$$

We must have $\bar{m}_k^E(1) \geq \bar{m}_k^E(2) \geq \dots \geq \bar{m}_k^E(N_R)$ for any user $k \in \mathcal{K}^E$. As the data rate corresponding to $\bar{m}_k^E(f)$ is $r^{\bar{m}_k^E(f)}$, the sum rate of f streams can be given by $f \cdot r^{\bar{m}_k^E(f)}$, if all streams are successfully received.

Finally, we determine the number of data streams $y_k(t)$ for each edge user by choosing an f that maximizes $f r^{\bar{m}_k^E(f)}$, i.e.,

$$y_k(t) = \arg \max_{f \leq N_R} f \cdot r^{\bar{m}_k^E(f)}.$$

Note that the determination of $y_k(t)$ is independent among the cell-edge users and therefore can be implemented in parallel.

Step 1-C: Determine Candidate MCS for Cell-Edge Users. In this step, we focus on $z_k^m(t)$ variables. In Step 1-B, we identified the largest MCS level $\bar{m}_k(y_k(t))$ based on the estimated SINR $\gamma_k^E(f)$. However, simply applying MCS $\bar{m}_k(y_k(t))$ to user k can be overly optimistic for a few reasons. First, $\gamma_k^E(f)$ is approximated based on channels' eigenvalues, while non-SVD based beamforming techniques, such as MMSE and ZF, cannot process signals in the eigenspace and will result in an inferior performance. Second, $\gamma_k^E(f)$ does not consider inter-cell interference. Third, $\gamma_k^E(f)$ is based on averaged eigenvalues over all $b \in \mathcal{B}_k^E$. Applying a lower MCS for user k has the potential to facilitate successful transmissions on more RBs and thus achieve a higher sum rate.

Therefore, instead of simply applying MCS $\bar{m}_k^E(y_k(t))$ to user k , we propose to picking up multiple MCS candidates that are lower than $\bar{m}_k^E(y_k(t))$ simultaneously. Then M^3 can process multiple MCS choices for a user in parallel. How to determine its final MCS jointly with all other users will be discussed in Stage III. Now let's focus on determining the MCS candidates for a particular cell-edge user k .

Note that any MCS that is much less than $\bar{m}_k^E(y_k(t))$ is not a promising candidate, as it can only support a low data rate. Therefore, M^3 only chooses a candidate MCS that is lower than $\bar{m}_k^E(y_k(t))$ but the difference between $\bar{m}_k^E(y_k(t))$ and this MCS is within a pre-defined value M_Δ . Formally, M^3 determines the candidate MCS set for user k by

$$\mathcal{M}_k^E = \{m \in \mathcal{M} | 0 \leq \bar{m}_k^E(y_k(t)) - m < M_\Delta\}.$$

After this step, we have $z_k^m(t) = 0$ if $k \in \mathcal{K}^E$ and $m \notin \mathcal{M}_k^E$. As described in problem OPT, the choice of a user's MCS is coupled with other (both cell-edge and non-edge) users. How to finalize MCS selection in $\mathcal{M}_k^E$ for all users will be discussed in Stage III (after we discuss MCS selection for non-edge users).

Pipeline 2: Non-Edge Users This pipeline is designed for non-edge users $\mathcal{K}^E$, assuming that each RB is available only for non-edge users. Note that for a user $k \in \mathcal{K}^E$, it receives a much higher signal strength from its own RRH than that from any other RRHs. Thus, when finding promising scheduling solutions at an RRH l , we can rely solely on the channel information within RRH l , while treating the inter-cell interference as noise. This allows M^3 's decision for non-edge users to be independent among different RRHs and therefore achieves a higher level of parallelism. Pipeline 2 consists of the following three steps.

Step 2-A: Select Promising MU-MIMO Users on Each RB under Each RRH. In this step, we deal with $x_{l,k}^b(t)$ variables for non-edge users $\mathcal{K}_l^E$. Compared to cell-edge users, non-edge users experience better SINR on average. To exploit this property for higher throughput, we employ MU-MIMO transmission for non-edge users, which is not used for cell-edge users. Under MU-MIMO transmission, more than one user will be selected on each RB under each RRH.

M^3 selects users on each RB for MU-MIMO transmission through two operations. First, M^3 finds a subset $\tilde{\mathcal{K}}_l^{b,E}$ of promising users from $\mathcal{K}_l^E$ based on channel quality. Second, M^3 further selects a subset $\mathcal{K}_l^{b,E}$ of promising users from $\tilde{\mathcal{K}}_l^{b,E}$ based on the channel correlation among users to form MU-MIMO transmission. We now describe each operation in detail.

First, M^3 intensifies promising MU-MIMO users on each RB based on channel quality. Similar to Pipeline 1, we introduce a metric $q_{l,k}^{b,E}$ to approximate user k 's data rate normalized by its long-term data rate, except that we now consider only one RRH for each non-edge user. Specifically, M^3 creates $\sum_{l \in \mathcal{L}} |\mathcal{B}| |\mathcal{K}_l^E|$ independent parallel flows, each of which calculates one $q_{l,k}^{b,E}$ for $l \in \mathcal{L}$, $k \in \mathcal{K}_l^E$ and $b \in \mathcal{B}$ and

$q_{l,k}^{b,E}$ is given by:

$$q_{l,k}^{b,E} = \frac{\log_2 \left(1 + \frac{P_T}{n_0^2} \|\mathbf{H}_{l,k}^b\|_F^2 \right)}{\tilde{R}_k(t-1)}.$$

Then, M^3 generates $|\mathcal{L}| |\mathcal{B}|$ independent parallel flows, each of which sorts $\{q_{l,1}^{b,E}, q_{l,2}^{b,E}, \dots, q_{l,|\mathcal{K}_l^E|}^{b,E}\}$ in descending order for a RRH $l \in \mathcal{L}$ and a RB $b \in \mathcal{B}$. Let $\pi_{l,k}^b$ be the order of $q_{l,k}^{b,E}$ in $\{q_{l,1}^{b,E}, q_{l,2}^{b,E}, \dots, q_{l,|\mathcal{K}_l^E|}^{b,E}\}$. Suppose M^3 selects K_Q ($< |\mathcal{K}_l^E|$) candidate users based on channel qualities, then the subset $\tilde{\mathcal{K}}_l^{b,E}$ of promising users is determined by

$$\tilde{\mathcal{K}}_l^{b,E} = \{k \in \mathcal{K}_l^E | \pi_{l,k}^b \leq K_Q\}. \quad (b \in \mathcal{B}, l \in \mathcal{L})$$

Second, we identify promising users from $\tilde{\mathcal{K}}_l^{b,E}$ on each RB to form MU-MIMO transmission based on channel correlations. That is, among the users in $\tilde{\mathcal{K}}_l^{b,E}$ for each $b \in \mathcal{B}$, $l \in \mathcal{L}$, M^3 selects a subset $\mathcal{K}_l^{b,E}$ ($\subset \tilde{\mathcal{K}}_l^{b,E}$) of users, such that the users in $\mathcal{K}_l^{b,E}$ have low channel correlations among themselves. The rationale behind this operation is that, in general, the lower correlations among the co-scheduled users, the higher sum of data rate can be achieved. This is because mutually orthogonal channels can better preserve the desired signal strength after applying beamforming matrices [23, 24].

We evaluate channel correlations based on *chordal distance* [25], which measures the angle between two multi-dimensional subspaces. A larger value of chordal distance means more orthogonality between these two subspaces. Let $\hat{\mathbf{H}}_{l,k}^b$ be the orthonormal base of $\mathbf{H}_{l,k}^b$. Then the chordal distance between $\mathbf{H}_{l,k_1}^b$ and $\mathbf{H}_{l,k_2}^b$ is given by:

$$d_l^b(k_1, k_2) = \frac{1}{\sqrt{2}} \|\hat{\mathbf{H}}_{l,k_1}^{b\dagger} \hat{\mathbf{H}}_{l,k_1}^b - \hat{\mathbf{H}}_{l,k_2}^{b\dagger} \hat{\mathbf{H}}_{l,k_2}^b\|_F.$$

M^3 computes $d_l^b(k_1, k_2)$ for all $k_1 \in \mathcal{K}_l^E$, $k_2 \in \mathcal{K}_l^E$, $k_1 \neq k_2$, which can be executed in parallel on each RB $b \in \mathcal{B}$ and $l \in \mathcal{L}$. Suppose we are going to select K_{MU} ($< |\mathcal{K}_l^E|$) users for MU-MIMO transmission. Then the subset $\mathcal{K}_l^{b,E}$ of candidate users is determined by the following. M^3 adds the first user to $\mathcal{K}_l^{b,E}$ that has the highest $q_{l,k}^{b,E}$, i.e., $\mathcal{K}_l^{b,E} = \{\arg \max_{k \in \mathcal{K}_l^E} q_{l,k}^{b,E}\}$. Subsequently, we add users one at a time to $\mathcal{K}_l^{b,E}$, by picking the user with the largest average chordal distance to existing users in $\mathcal{K}_l^{b,E}$, until we have K_{MU} users in $\mathcal{K}_l^{b,E}$.

After Step 2-A, M^3 restricts the users that can be scheduled on RB b under RRH l . That is, for any $k \in \mathcal{K}_l^E$ and $k \notin \mathcal{K}_l^{b,E}$, we have $x_{l,k}^b(t) = 0$. On the other word, under each RRH and each RB, the total number of possibilities to allocate RBs is reduced to $2^{K_{\text{MU}}}$, as $x_{l,k}^b(t)$ can be either 1 or 0 and $|\mathcal{K}_l^{b,E}| = K_{\text{MU}}$. The final decision will be made in Stage III.

Step 2-B: Determine Number of Data Streams for Non-edge Users. In this step, we work on $y_k(t)$ variables for non-edge users. Similar to cell-edge users, we consider a metric of estimated SINR as a function of the number of streams. But for a non-edge user, we only need to consider its own RRH.

That is, if RRH l 's non-edge user k is receiving f streams, the estimated SINR per stream is given by

$$\gamma_{l,k}^E(f) = \frac{P_T}{f} \frac{E_b[\sigma_{l,k}^b(f)]^2}{n_0^2},$$

where $E_b[\sigma_{l,k}^b(f)]$ is the average eigenvalue over all $b \in \mathcal{B}_{l,k}^E$, and $\mathcal{B}_{l,k}^E$ is the set of candidate RBs for non-edge user k under RRH l , i.e., $\mathcal{B}_{l,k}^E = \{b \in \mathcal{B} | k \in \mathcal{K}_{l,k}^E\}$.

Next, following the same rationale as cell-edge users, let $\bar{m}_k^E(f)$ be the largest MCS that satisfies $\gamma_{l,k}^E \geq \theta^{\bar{m}_k^E(f)}$. Then the number of data streams $y_k(t)$ for a non-edge user k is determined by choosing the f that maximizes $f \cdot r^{\bar{m}_k^E(f)}$, i.e.,

$$y_k(t) = \arg \max_{f \leq N_R} f \cdot r^{\bar{m}_k^E(f)}.$$

Step 2-C: Determine Candidate MCS for Non-edge Users.

The MCS $\bar{m}_k^E(y_k(t))$ found in Step 2-B for a non-edge user is also an optimistic choice, because the transmit power P_T shall be shared with other non-edge users due to the MU-MIMO transmission. Similar to the approach for cell-edge users, M^3 only chooses a candidate MCS that is lower than $\bar{m}_k^E(y_k(t))$ and the difference between $\bar{m}_k^E(y_k(t))$ and this MCS is within a pre-defined value M_Δ . That is, the candidate MCS set for non-edge user k is given by

$$\mathcal{M}_k^E = \{m \in \mathcal{M} | 0 \leq \bar{m}_k^E(y_k(t)) - m < M_\Delta\}.$$

After this step, we have $z_k^m(t) = 0$ if $m \notin \mathcal{M}_k^E$ for a non-edge user k .

C. Stage III: Determine final solution

In this stage, M^3 performs comparisons among different candidate solutions and selects the final solution to problem OPT.

M^3 will enumerate all possible $x_{l,k}^b(t)$ and $y_k(t)$ assignments from each pipeline in Stage II. However, for $z_k^m(t)$ variables, the search space is too large for enumeration. Fortunately, the MCS candidates $\mathcal{M}_k^E$ and $\mathcal{M}_k^E$ for cell-edge and non-edge users (identified by Stage II) are already promising MCS solutions. Therefore, M^3 can further identify a smaller subset of MCS solution space. Specifically, let $\tilde{\mathcal{M}}$ be the MCS solution space, i.e., $\tilde{\mathcal{M}} = \mathcal{M}_1^E \times \dots \times \mathcal{M}_{|\mathcal{K}^E|}^E \times \mathcal{M}_1^E \times \dots \times \mathcal{M}_{|\mathcal{K}^E|}^E \subseteq \mathcal{M}^{|\mathcal{K}|}$, where $\times$ denotes the Cartesian product. Then each element from $\tilde{\mathcal{M}}$ (a $1 \times |\mathcal{K}|$ vector) is a feasible MCS solution for all users. Supposing we are going to try M_S different MCS solutions in parallel, we can apply a simple yet effective approach to determine MCS solutions—by randomly (uniform) picking M_S elements from $\tilde{\mathcal{M}}$. After this step, we have M_S candidate MCS solutions (i.e., M_S sets of feasible $z_k^m(t)$'s), denoted by $\mathcal{Z}_1, \mathcal{Z}_2, \dots, \mathcal{Z}_{M_S}$, respectively.

After obtaining $\mathcal{Z}_i$'s, M^3 calculates SINR of each data stream and PF metrics under a given $z_k^m(t)$. Specifically, M^3 enumerates all possible $x_{l,k}^b(t)$ and $y_k(t)$ assignments for cell-edge and non-edge users from Stage II. With the given

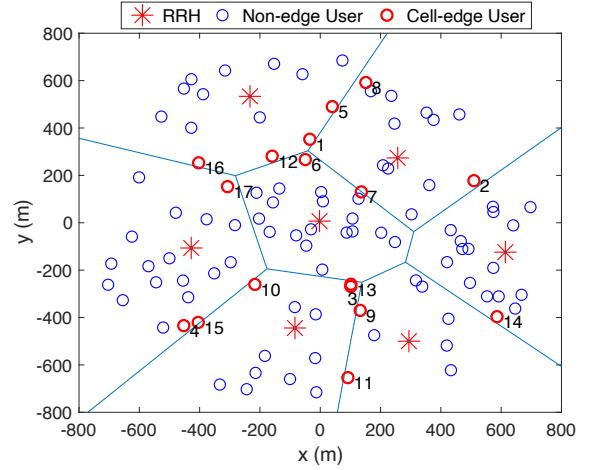


Fig. 5: An instance of network topology with 7 RRHs and 100 users.

$x_{l,k}^b(t)$'s and $y_k(t)$'s, we derive the corresponding beamforming matrices (based on MMSE beamforming) at each RRH and each user, and then calculate SINR for every stream. When calculating user k 's SINR, we exploit independency and reduce the computational burden by only considering the beamforming gain from user k 's RRHs $\mathcal{L}_k$. That is, for interference from RRHs other than $\mathcal{L}_k$, we only consider their transmit power P_T and pathloss attenuation, without using their beamforming matrices to obtain the inter-cell interference.² Although such a simplification has little impact on SINR accuracy, it can decouple SINR calculations among the RRHs and make it possible for parallel implementation.

Next, with given SINRs and $z_k^m(t)$'s, we can obtain PF metric for all the candidate assignments in parallel. Then M^3 compares the PF objective values achieved by each candidate solution (a combination of $x_{l,k}^b(t)$'s, $y_k(t)$'s and $z_k^m(t)$'s). The independency among RBs and parallelism can be exploited for the comparison. We omit the details to conserve space. Finally, the best solution that provides the highest PF metric will be chosen as the final solution to problem OPT.

V. IMPLEMENTATION AND EXPERIMENTAL EVALUATION

A. Implementation

The exploration of independent operations in our design in Sec. IV makes it possible for a parallel implementation. We implement M^3 on an NVIDIA DGX station, which comes with 4 V100 GPU cards. We will only use two of them. Each V100 card has 5120 CUDA cores. Data communication between CPU and GPU is based on a PCIe V3.0 architecture, and the GPU-to-GPU data communication is based on NVIDIA NVLink architecture [26]. The CPU of DGX Station is Intel Xeon E5-2698 v4 2.2 GHz (20-core). The programming platform is CUDA v10.2 [27]. Further, discreet engineering efforts are made to optimize timing performance, such as minimizing data exchange between pipelines / GPU

²All beamforming matrices will be considered for throughput evaluation in Sec. V.

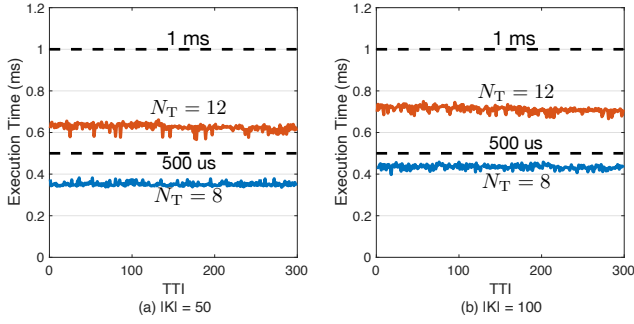


Fig. 6: M^3 's execution time. (a) $|\mathcal{K}| = 50$, $N_T \in \{8, 12\}$, and (b) $|\mathcal{K}| = 100$, $N_T \in \{8, 12\}$.

cards and managing data transfer between host and GPU (the data transfer time will be counted toward total execution time). We omit the implementation details to conserve space.

B. Settings

Considering a C-RAN architecture, we randomly generate $|\mathcal{L}| = 7$ RRHs in a circle within a radius of 700 m. The minimum distance between every two RRHs is 350 m. $|\mathcal{K}|$ users are randomly deployed in the circle, and $|\mathcal{K}| = 100$ unless stated otherwise. Fig. 5 shows an instance of the network topology and the results of user classification done by Stage I. In Fig. 5, the red circles are the cell-edge users and the blue circles are non-edge users. Fig. 5 shows that we have 17 cell-edge users (out of 100 users) in this case ($\delta = 3$ dB).

The number of antennas at each RRH is chosen from $\{8, 12\}$ and the number of antennas at each user is 2. The number of MU-MIMO users K_{MU} is 2 or 4. The number of RBs $|\mathcal{B}|$ is 100. For the wireless channels, $\mathbf{H}_{l,k}^b(t)$ includes both large-scale fading $g_{l,k}$ and small-scale fading $\bar{\mathbf{H}}_{l,k}^b(t)$, i.e., $\mathbf{H}_{l,k}^b(t) = g_{l,k}^{-1} \bar{\mathbf{H}}_{l,k}^b(t)$. Large-scale fading $g_{l,k}$ is given by $140.7 + 36.7 \log_{10}(d_{l,k})$ (in dB), where $d_{l,k}$ is the distance between RRH l and user k (in km). The small-scale fading $\bar{\mathbf{H}}_{l,k}^b(t)$ is modeled by Rayleigh channel model. We set transmit power P_T to be 36 dBm. The background noise power is set to -169 dBm/Hz and the channel bandwidth is 20 MHz. For parameters δ , K_Q , M_Δ , and M_S in our algorithm (see Sec. IV), we set $\delta = 3$ dB, $K_Q = 10$, $M_\Delta = 6$, $M_S = 300$.

C. Timing Performance

We first verify that M^3 can meet the 1 ms real-time requirement, which is a major criterion for it to be useful for 5G C-RAN. We conduct experiments for 300 consecutive TTIs under two different settings: (a) $|\mathcal{K}| = 50$, $N_T \in \{8, 12\}$, and (b) $|\mathcal{K}| = 100$, $N_T \in \{8, 12\}$. The experimental results are shown in Fig. 6. We find that that M^3 is able to offer a scheduling solution within 1 ms under all cases and TTIs in our experiments. Specifically, when $N_T = 12$, M^3 's average execution time is 626 μs and 712 μs for $|\mathcal{K}| = 50$ and $|\mathcal{K}| = 100$, respectively, which can meet the timing requirement for 5G NR numerology 0 (1 ms). When $N_T = 8$, the average execution time is 351 μs and 435 μs for $|\mathcal{K}| = 50$ and $|\mathcal{K}| = 100$, respectively, which can meet 5G NR numerology 1 requirement (500 μs).

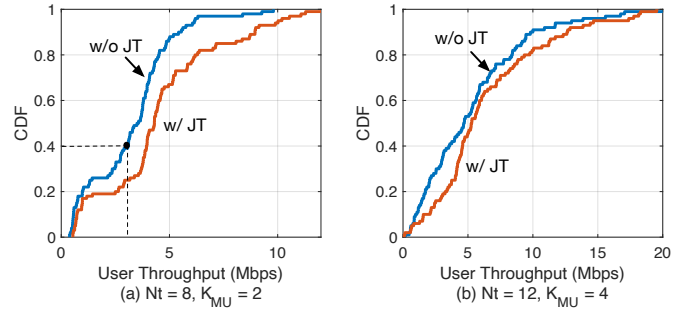


Fig. 7: Comparison of CDFs of users' long-term average throughput.

D. Throughput Performance

We now evaluate M^3 's throughput performance. Under the topology in Fig. 5, we compare the throughput performance achieved under joint transmission (with $\delta = 3$ dB) with the case when joint transmission is not used (i.e., $\delta = 0$ dB). We consider two different settings: (a) $N_T = 8$, $K_{\text{MU}} = 2$, and (b) $N_T = 12$, $K_{\text{MU}} = 4$.

Fig. 7 shows the cumulative distribution functions (CDF) of users' long-term average throughput. For example, in Fig. 7(a), the point (3.09 Mbps, 0.4) on the blue curve indicates that the 40th lowest user throughput (among 100 users) is 3.09 Mbps. The results in Fig. 7 suggest that the design of M^3 is able to offer a better throughput performance when joint transmission (JT) is employed. Specifically, we examined user throughput at different percentiles ranging from 5th to 40th percentile with and without joint transmission. The results suggest that the user throughput is increased by at least 20% at all examined percentiles and can be up to 160% for some cases. On average (over 100 users), the user throughput is improved by 40.5% and 20.9% through joint transmission under settings (a) and (b), respectively.

VI. CONCLUSIONS

This paper presented M^3 —the first real-time scheduler for a multi-cell MIMO system under C-RAN architecture. M^3 jointly optimizes RB allocation, MCS assignment, and beamforming matrices for all users under all RRHs and is able to offer a solution within 1 ms. To address the stringent real-time requirement, we developed a novel multi-pipeline design that exploits large-scale parallelism. For validation, we implemented M^3 on a COTS Nvidia DGX Station. Through extensive experiments, we showed that M^3 can find a scheduling solution within 1 ms for all tested cases, while it can significantly increase user throughput by leveraging joint transmission among neighboring cells.

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